THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS CHAPTER TEST SET A

Chapter 5: **Prime Time** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A number having exactly two factors, 1 and itself, is: 1

(a) an even number **(b)** a composite number
(c) a prime number **(d)** a co-prime

2. Which of these is a composite number? 1

(a) 59 **(b)** 47 **(c)** 53 **(d)** 51

3. The method used to find all primes up to 100 by crossing out multiples is the: 1

(a) Sieve of Eratosthenes **(b)** Kaprekar routine
(c) Collatz routine **(d)** Pascal's method

4. A number is divisible by 5 if its last digit is: 1

(a) 2 or 5 **(b)** 5 or 7 **(c)** 0 or 5 **(d)** 1 or 5

5. The smallest prime number is: 1

(a) 2 **(b)** 3 **(c)** 0 **(d)** 1

6. Every number is a multiple of: 1

(a) 2 **(b)** 10 **(c)** 1 and itself **(d)** its square

7. Prime factors of 30 are: 1

(a) 2, 3, 6 **(b)** 3, 10 **(c)** 2, 3, 5 **(d)** 1, 2, 15

8. The factors of a number are always: 1

(a) primes only **(b)** greater than the number
(c) odd only **(d)** less than or equal to the number

9. The HCF of 14 and 21 is: 1

(a) 7 **(b)** 8 **(c)** 14 **(d)** 6

10. The HCF of 10 and 35 is: 1

(a) 5 **(b)** 6 **(c)** 10 **(d)** 4

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Define a prime number. Give three examples. 2

12. List the factors of 24. 2

13. Write the first five multiples of 7. 2

14. Find the HCF of 18 and 24. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Find the prime factorisation of 84 and 126 and their HCF. 3

16. Explain the Sieve of Eratosthenes in steps. 3

17. Find the LCM of 12 and 18 using prime factorisation. 3

OR

Using divisibility tests, check whether 3,456 is divisible by 2, 3, 4, 6 and 9. 3

18. Write all primes less than 30 and state how many there are. 3

OR

Explain with an example why two consecutive numbers are always co-prime. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Find the prime factorisation of 180. 5
(ii) Find its number of factors.
(iii) List all its factors.

OR

(i) Find the HCF and LCM of 24 and 36 by prime factorisation. 5
(ii) Verify HCF × LCM = product.

20. (i) State the divisibility tests for 2, 3, 4, 5, 6, 9, 10. 5
(ii) Test 5,940.

OR

Case study: A teacher has 36 pencils and 48 erasers to pack equally into the largest number of kits without leftovers. 5
(i) How many kits?
(ii) Pencils in each?
(iii) Erasers in each?
(iv) Which concept is used?

END OF QUESTION PAPER Total: 40 Marks

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